

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An autonomous Institution affiliated to Anna University)

Degree & Branch	BE Computer Science & Engineering	Semester	V
Subject Code & Name	UCS3512 & Principles of Machine Learning Laboratory		
Academic year	2026-2027 (Odd)	Batch:2023-2027	Due date: 5.7.2026

Assignment 1: Working with Python packages-Numpy, Scipy, Scikit-Learn, Matplotlib

1. Explore the various functions and methods provided by the following Python libraries. [CO1, K3]
NumPy – Array creation, indexing, slicing, reshaping, mathematical operations, and statistical functions.
Pandas – DataFrame and Series creation, data loading, indexing, filtering, handling missing values, and descriptive statistics.
SciPy – Mathematical and scientific computing functions, optimization, statistics, and linear algebra.
Scikit-learn – Dataset loading, data preprocessing, train-test splitting, model training, prediction, and performance evaluation.
Matplotlib – Creating basic plots such as line charts, bar charts, scatter plots, histograms, and customizing visualizations.
Demonstrate the use of selected functions from each library with suitable Python examples.
2. Explore public repositories such as the UCI Machine Learning Repository and Kaggle. Download the following datasets and identify the appropriate machine learning paradigm (Supervised, Unsupervised, or Semi-supervised) and the type of task performed (Classification or Regression). [CO1, K3]
 - i. Loan Amount Prediction
 - ii. Handwritten Character Recognition (MNIST)
 - iii. Email Spam Classification
 - iv. Diabetes Prediction
 - v. Iris Dataset
 - vi. Mall Customer Segmentation

Identify the type of task performed for above datasets and tabulate the details in the below table.

Dataset	Machine Learning Paradigm	Classification/Regression/Clustering	Target Variable
Iris Dataset			
Loan Amount Prediction			
Diabetes Prediction			
Email Spam Classification			
Handwritten Character Recognition (MNIST)			
Mall Customer Segmentation			

3. Explore the steps involved in the machine learning workflow and explain it using appropriate libraries and methods [CO1, K2].
 - a. Download any one of the following datasets:

- Loan Amount Prediction
 - MNIST
 - Email Spam Classification
 - Diabetes Prediction
 - Iris Dataset
- b. Load the dataset into Python using the Pandas library.
 - c. Display the first five records of the dataset.
 - d. Determine: Number of rows, Number of columns, Feature names, Target (label) column
 - e. **Data Preprocessing:**
 - Identify and handle missing values- Explain the method used to handle missing values.
 - Remove irrelevant features - Justify whether they should be removed.
 - Encode categorical variables - Justify the technique used for encoding
 - Normalize or standardize numerical features.
 - f. Compute the Correlation Coefficient and Mutual Information scores for all input features with respect to the target variable. Tabulate the results and identify the top 'n' important features.
 - g. Exploratory Data Analysis (EDA) and Visualization: Examine the dataset using summary statistics and visualize the data using plots such as histograms, bar charts, scatter plots, box plots, heatmaps, and pair plots.
 - h. Based on the visualizations, answer the following:
 - Which features appear normally distributed?
 - Is the dataset balanced or imbalanced (for classification datasets using any chart)?
 - i. Data Splitting: Divide the dataset into training, validation, and testing sets for model development and evaluation.

Note:

A supplementary document on Exploratory Data Analysis (EDA) has been uploaded to the LMS portal to facilitate a better understanding of the concepts and visualizations covered in this experiment.

Optional Task:

Explore the features, installation procedures, and basic functionalities of popular deep learning frameworks such as TensorFlow, Keras, and PyTorch. Compare their key characteristics, applications, and typical use cases in deep learning.

Submission Guidelines

Observation Note

State the aim of the experiment, record the tasks performed and corresponding observations, summarize the key learning outcomes.

LMS Submission (Single PDF)

The submitted report should include:

- Aim of the experiment.
- Python code segments with appropriate comments.
- Output screenshots for the implemented tasks.
- Summary of observations/findings in a tabular format.
- Inference based on the experiment and reflection on the learning outcomes.